

Capstone Proposal

Overview :

The News Summarizer fetches daily tech news from NewsAPI's free tier, uses AWS Bedrock to whip up concise, AI-powered summaries, and stores the polished articles in DynamoDB. It runs on a schedule with EventBridge, which kicks off an ECS Fargate task to pull the articles, summarize them, and save everything for future reference.

Behind the scenes, the setup relies on ECS Fargate for containerized compute (no need to fiddle with EC2), ECR for the Docker image, Secrets Manager to keep the NewsAPI key safe, DynamoDB for keeping state, and CloudWatch for logs and monitoring. In short, raw news gets transformed into curated, bite-sized summaries and built up over time, thanks to both data enrichment and a bit of temporal tracking.

Services:

Service	Role	What It Does
ECS Fargate	Compute	Runs your containerized Python app that fetches & summarizes news
ECR	Container Registry	Stores your Docker image
EventBridge	Trigger/Scheduler	Runs the ECS task every 24 hours
DynamoDB	Persistent State	Stores summarized articles with dates
Secrets Manager	Secret Management	Safely stores NewsAPI key
CloudWatch	Observability	Logs, metrics, monitoring

Architecture Diagram :

